

US Stock Time-Series Forecasting

Business Problem:

As a financial analyst, we need to develop quality models in order to predict future price movement. This will allow us to more accurately give advice to our customers on when to invest their hard-earned money in the market. It can be difficult to advise customers because of the wave like nature of stock price movements in addition to other factors. Can we develop a model that will accurately suggest where the bottom of the waves will be.

Background:

Financial analysis is the process of evaluating a company's financial performance and health. It involves examining financial statements, ratios, and other metrics to gain insights into the company's key indicators. Financial analysis is a crucial tool for investors, creditors, and other stakeholders to make informed decisions about the company's future (History 2019). Financial analysis as a trading mechanism has been around in some form for centuries, but the modern idea of it took off in the early 20th century. With modern computing systems, financial analysts can run seemingly limitless factors through models in order to predict the best points to invest in a company's stock. These financial analysts are relied upon by many in the general populace to recommend where to put their money and when to do it.

One of the best/most common ways to make predictions about the future is to study the past. This method is known as Time series analysis. There are many methods to develop a time series analysis model, but one of the prominent ways to do so was developed by Facebook's data scientist called Prophet. Many of the other models struggle with effects caused by human behavior, such as seasonal effects, trends or outliers. Prophet was developed to combat these issues. It was released in 2017 as open source

software (Kashnitsky, 2020). Using prophet for this project task would be a good choice because US stocks are heavily subjected to the whims of human behavior and trends over the course of the year.

Data Explanation:

The data for this project was obtained from a public data directory, Kaggle.com. The data consists of a vast amount of US stock market tickers. Multiple key dollar values were recorded from early 2013 until early 2018. Because of how Prophet functions each ticker would have to be run individually. For time and efficiency's sake, I chose to pick one ticker that I knew would be subject to seasonal shifts. The American Airlines ticker was chosen because the population increases flight traffic based on summer vacation or winter holidays with the time in between showing limited flight traffic.

Methods:

Data cleaning and analysis was all performed through Jupyter Notebooks using various Python libraries. The data was first cleaned and assessed for any null values. The 'date' column was specifically converted to datetime formatting, as required by the Prophet library. I then decided to focus on the close price for the ticker. This seemed the best course because there can be significant volatility for the price during the day, but the closing price usually will follow a trend as long as nothing drastic happens.

Analysis:

Very little data cleaning was required. Because of this, the data could be run through Prophet very quickly. Figure 1 shows the predicted forecast for the stock price for almost a year after the last data point was logged. The thick blue area represents the possibility that the price is predicted to go anywhere in that area. The dark blue area is where the higher probability is. A trend can be seen with this chart, however the trend is easier seen in Figures 3 and 4. The chart has been smoothed and focused, reducing the noise from the first figure. The model suggests that the trend for the American Airlines stock price is to increase over the next year after early 2018. One of the key investment mantras is to invest low and sell high. With

just these figures, we can see that early 2018 is a good time to buy the stock based on predictions. Using Figure 2, we can see the seasonality for the stock price. The price jumps during vacation times in the summer and holiday travel times in the winter. Knowing this, we can recommend buying into the stock at other times of the year. It can also be used to show investors that they shouldn't be worried during the September price slump because a price boom should be close in the future with the holidays. The moving average is an analysis tool that smooths out the volatility in the price of a stock. It is mainly used by investors to find the trends in the prices. Figures 5-7 confirm the trend seen early in the analysis that the stock price is steadily increasing over time.

Conclusions:

The analysis for this project showed key insights into the changes for the American Airlines stock price. Prophet was the backbone of the analysis that provided quality calculations that accounted for the seasonality of air travel that would impact our data. Using this model, we can provide a decent recommendation for when to invest in this particular stock in order to make money from the investment. This model can be the blueprint for assessing any stock ticker that would be of interest to a potential client. It should be noted that this model is just a prediction and can only give limited suggestions. It will not be able to account for any drastic changes that ripple through the entire market, also known as 'black swan' events (Scott, 2024).

Assumptions:

The main assumption in this project is that the statistical properties of the data are consistent over the course of the period studied.

Limitations:

As always, time limitations were a factor.

Challenges:

There were difficulties with Prophet and its dependencies. It is not always the easiest library to work with. Without those difficulties, I would have analyzed more tickers.

Future Uses:

Modeling this type of data will greatly benefit many individuals and companies. Using machine learning as a tool to help improve financial analysis recommendations is of great benefit to the population. Better recommendations allow the 'mom and pop' investor to make better decisions and reach their financial goals.

Recommendations:

Other methods should be tested that can build time series models. Again, prophet isn't always the easiest to use.

Implementation Plan:

There are no implementation plans as of yet, but this model could be a key component to a broader collection of models used for financial analysis.

Ethical Assessment:

As financial advisors, we have a code of ethics to uphold. There shouldn't be any conflicts of interest between us and our customers. We are here to advise, not sell something. We should strive to understand our customers circumstances before providing advice. We have to report any signs of stock manipulation or financial abuse. Analyzing the US stock market can be difficult. It is not always stable. This is especially true today when there is such turmoil in the culture of the US. More models need to be used than this one in order to make an ethical recommendation to people about investing their money.

Reference List:

The Complete Guide to Time Series models. Built In. (n.d.). <https://builtin.com/data-science/time-series-model>

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<https://www.fintibi.com/history-and-the-development-of-financial-analysis/>

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Scott, G. (2024, June 5). *Black swan in the stock market: What is it, with examples and history*. Investopedia.
<https://www.investopedia.com/terms/b/blackswan.asp#:~:text=A%20black%20swan%20is%20an,t hey%20were%20obvious%20in%20hindsight.>

Appendix:

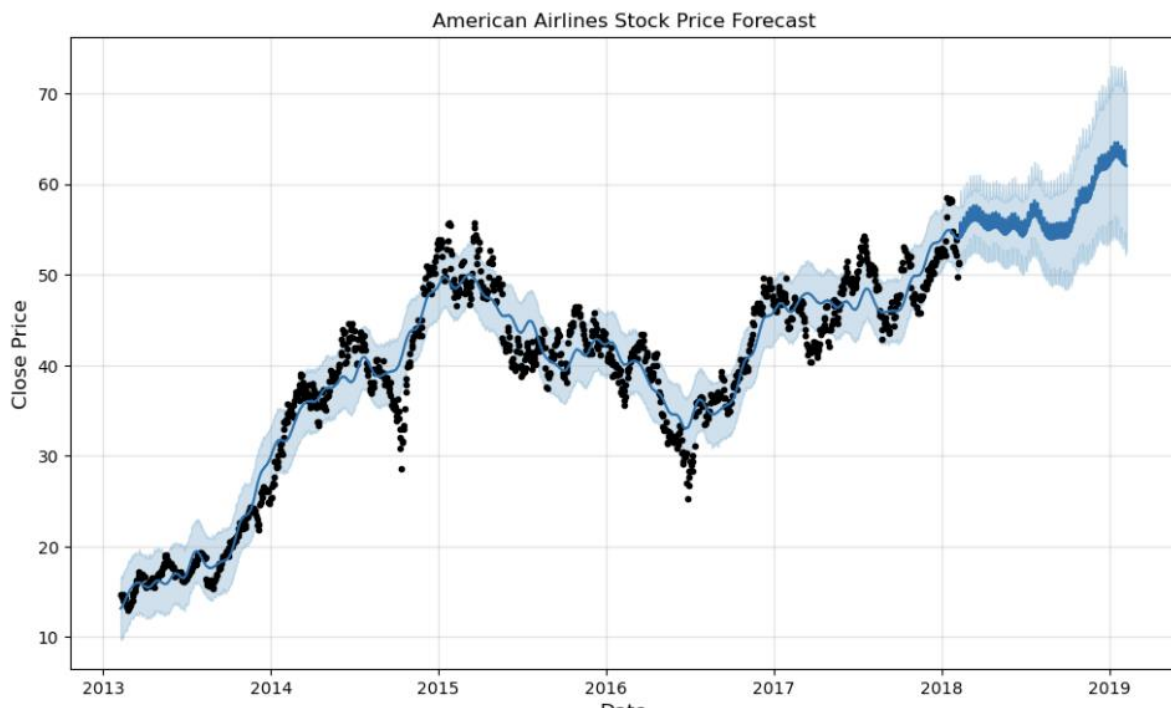


Figure 1 - AAL Stock Forecast

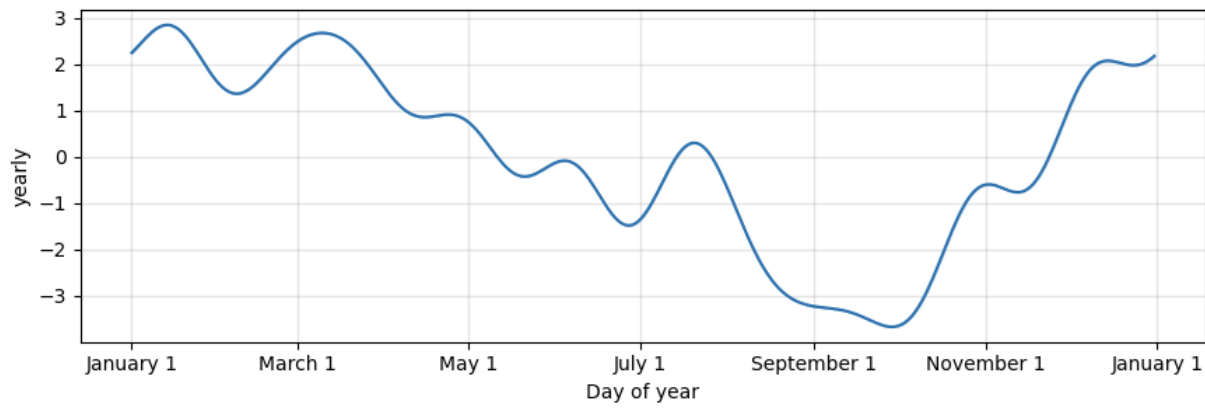


Figure 2 - AAL Seasonality

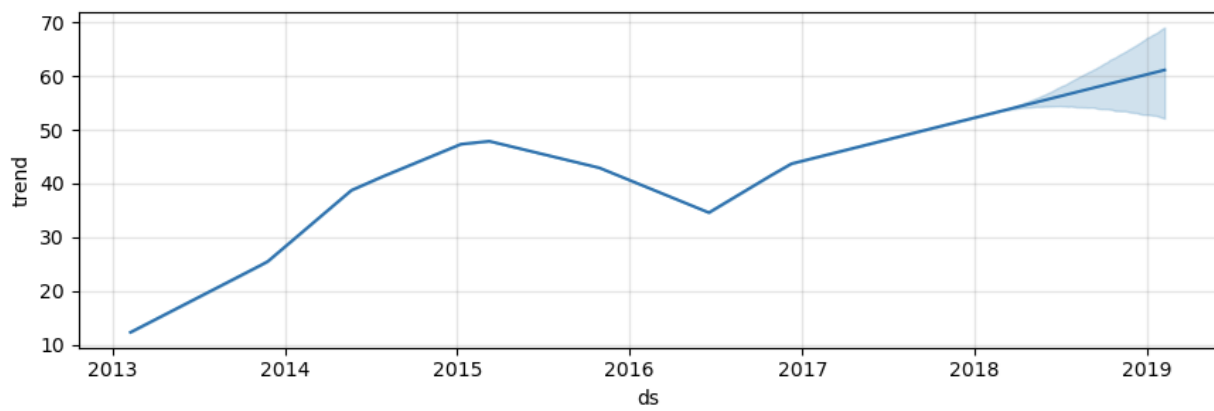


Figure 3 - AAL Stock Trend

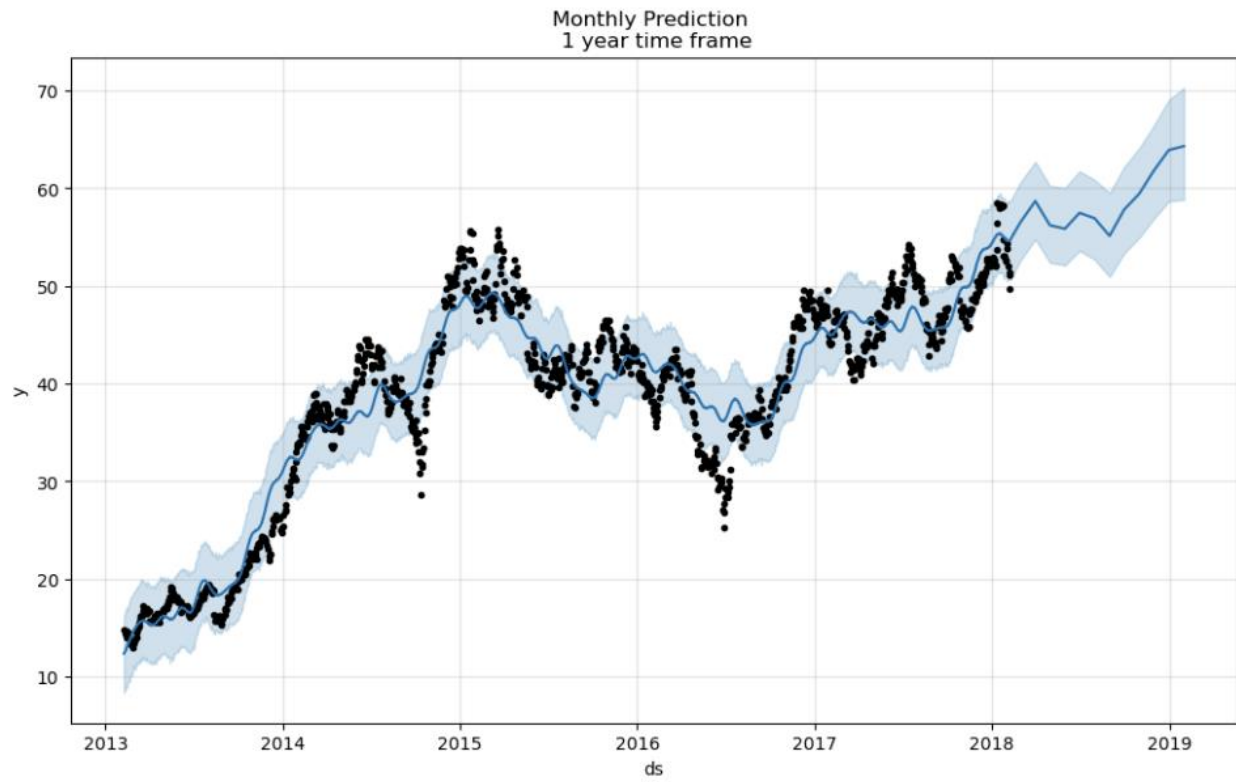


Figure 4 - Smoothed AAL Stock Forecast

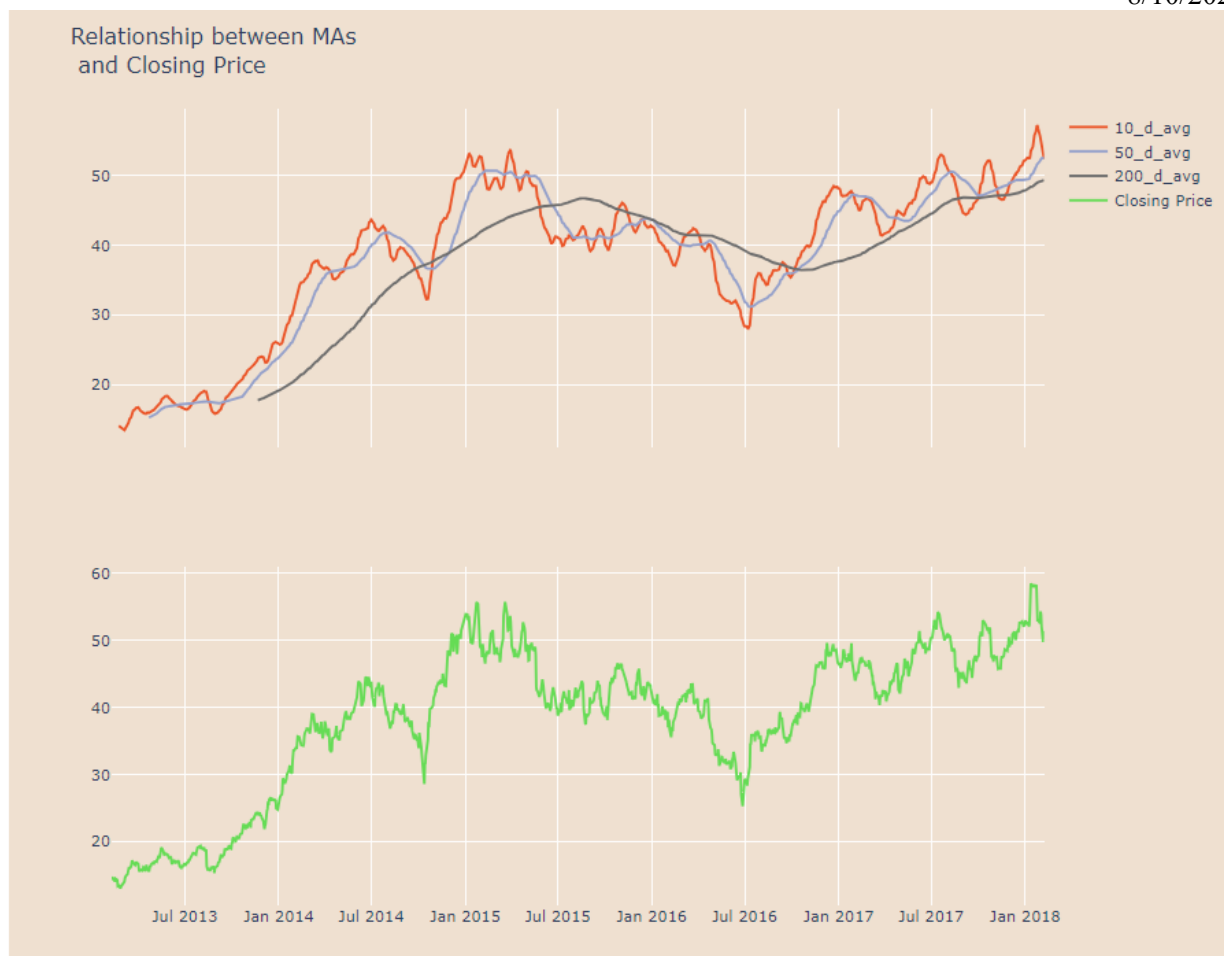


Figure 5 - Closing Price relationship to the Moving Averages

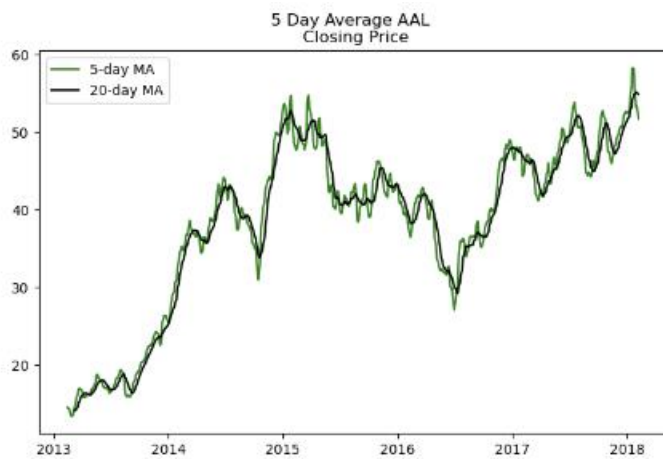


Figure 6 - Comparison of 5 and 20 day Moving Average

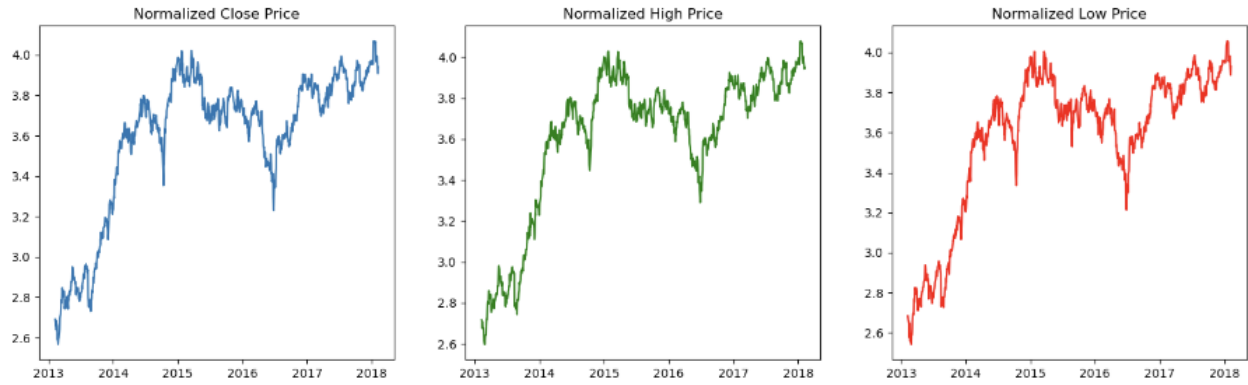


Figure 7 - Normalized Price Values

10 Questions:

1. Is time series analysis a good model for US Stocks?

Yes, it should be a part of a combination of models for stocks.

2. Would you use Prophet again?

Yes I would, but I hope they fix the bugs with the current version.

3. What other time series models are available?

Moving average, exponential smoothing, double exponential smoothing, triple exponential smoothing, ARIMA, SARIMA

4. What are the ethical considerations for financial analysis?

The main one is to not have a conflict of interest when providing stock recommendations. There are a lot of regulations on financial advisors and what stocks they can hold and recommend.

5. Is time series analysis enough to predict stock prices?

No, it should be a part of a multivariate modeling kit

6. Should the financial advisor/company be responsible for people losing money?

If it is found that there is a conflict of interest during the financial advising, then yes the advisor/company should be responsible for people losing money.

7. What other factors would help predict stock prices?

Sentiment analysis would be one thing. Many stocks are subject to trend changes. This can cause a lot of volatility in the price.

8. Would you be comfortable giving financial advice?

No, the potential for people losing all their saved money because of the poor model is too much for me.

9. What is the difference between ARIMA and SARIMA models?

SARIMA incorporates seasonality into the ARIMA model. This would have made SARIMA useful as a second model in this project.

10. Explain the concept of stationarity in time series analysis.

Stationarity is related to the statistical properties of time series data. If the mean and variance of the data is constant, then the data is stationary. This project involved data that was not stationary. Stocks tend to have high changes to the variance of their prices.